

Emre CANCIOGLU

DIGITALIZATION & AI SUPERVISOR · INDUSTRIAL AUTOMATION

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9+ Years Experience | 3 Facilities · 2-Person Team | 57% Cost Reduction | 4 Publications + 6 Awards

Summary

Results-driven Digitalization & AI Supervisor with over 9 years of experience leading digital transformation and applied-AI initiatives on the industrial production floor. Currently overseeing OT digitalization and AI strategy across **3 manufacturing facilities**, managing a 2-person engineering team, coordinating across **10+ cross-functional departments**, and leading the company-wide **AI Ambassador** program. Delivers measurable impact through **predictive maintenance**, **computer-vision-based safety systems** (contributing to **4 consecutive years of zero workplace accidents** in the areas where they were deployed), **GenAI-powered automation**, and **energy & sustainability analytics**, consistently translating shop-floor data into operational efficiency, cost savings, and safer working conditions. Deep hands-on expertise with **PLC, SCADA**, and industrial protocols (**OPC UA, MQTT, Profinet**), balanced with strong software/cloud and leadership capability (**Node.js, React, Azure/AWS, Kubernetes**). Recognized with multiple 1st-place finishes and an Innovation Special Award across İnci Holding's group companies. Also the author of **4 separate scientific publications** in applied machine learning and industrial automation.

Skills

OT Systems	PLC (Siemens, Allen-Bradley, Mitsubishi) SCADA DCS HMI MES OPC UA MQTT Modbus (TCP/RTU) Profinet EtherNet/IP CC-Link Kepware Unified Namespace Digital Twin ISA-95 NAT VLAN Segmentation Industrial Firewalls IEC 62443
AI & Data	Predictive Maintenance Computer Vision Machine Learning Deep Learning GenAI/LLM Automation (RAG, n8n, Copilot, GPT, Claude) MLOps Edge Computing ETL/Stream Processing SPC Grafana Power BI Data Analytics
Leadership	Team Leadership & Mentoring Cross-functional Collaboration Project & Budget Management Bias for Action
DevOps	Azure AWS Docker Kubernetes Rancher Terraform Jenkins GitHub Actions NGINX MS IIS PM2 Git
Development	Node.js Express.js Django FastAPI REST API Kafka React Redux JavaScript TypeScript Python MS SQL MongoDB PostgreSQL DynamoDB Elasticsearch
Languages	Turkish (Native) English (Professional)

Work Experiences

İNCİ GS YUASA/İNCİ AKÜ — MANİSA, TÜRKİYE

DIGITALIZATION & AI SUPERVISOR

Jan 2026 - Present

- Leading OT digitalization and applied-AI strategy across **3 production facilities**; managing a 2-person engineering team and coordinating across **10+** cross-functional departments. Delivered **15+** process-improvement projects since taking on the role.
- Scaled the OT data integration backbone to **300+** managed data channels and database connections (up from ~200) across **PLC, SCADA, MES**, and historian systems — integrating **36** new production lines/machines in 2026 alone (furnaces, mills, assembly, filling, and welding stations), surpassing the year's **20%** connectivity target to reach **25.76%** facility-wide machine coverage, and building a self-service **OPC UA**-based alerting platform that lets any user configure real-time custom alarms across all connected machines.
- Grew the applied-AI portfolio to **4** active predictive-maintenance/quality models and **25** AI-driven automation projects, including RAG-based knowledge systems and GenAI/multi-agent workflows (**n8n**, Microsoft Copilot, GPT, Claude) that cut day-long reporting cycles down to seconds; leading the **AI Ambassador (AI Elçileri)** program, driving every company department to build and own its own AI agent.
- Deployed real-time air-quality/particulate monitoring in areas with high lead-dust exposure, contributing to a **58.62%** reduction in workers' blood lead levels; also added real-time anomaly detection on production/utility systems that helped prevent a potential fire incident.
- Integrated two **YOLO**-based **computer-vision** safety models: **AI-ODS**, which auto-stops robots when a person enters their work envelope, and **AI-FDS**, which uses real-time light-projection warnings to prevent forklift-pedestrian and forklift-forklift collisions at blind spots — sustaining **zero workplace accidents for 4 consecutive years** in the areas where these systems were deployed.
- Retrofitted **200+** electricity, water, and natural-gas meters to remote-reading, built the underlying database, and launched an energy dashboard visualizing historical machine-level consumption, per-unit-product energy use, and carbon footprint — integrated with **SAP** and preparing the facility for **ISO 50001** certification.
- Built a compressed-air leak anomaly-detection algorithm fusing compressor pressure data, real-time **MES** production signals, and **SAP** machine-level production-confirmation data, enabling early leak intervention and reducing compressor energy consumption.
- Added ML-based energy forecasting driven by planned production volumes (**XGBoost, LightGBM, GBM** regression, R^2 up to **0.944**), co-authored as peer-reviewed research at the 13th International European Conference on Interdisciplinary Scientific Research (Tirana, Albania, May 2026).

- Architected a **Unified Namespace (UNS)** unifying **SAP, MES, and PLC/SCADA/DCS**-based shop-floor equipment — communicating over **OPC UA, MQTT, Kafka, Profinet, EtherNet/IP, and CC-Link** — with energy analyzers and third-party systems, establishing the facility's digital twin; led a **20-person** team to give every battery full-lifecycle process and test traceability from raw material to shipment.
- Led the design and rollout of the OT network architecture; ran **80+** structured cybersecurity and disaster-recovery operations across **VLAN** segmentation, **NAT**, and **industrial firewalls** aligned to the **ISA-95/Purdue** model and **IEC 62443** to safeguard production continuity and data security.
- Built a **46-channel, 156-database** backbone (**MS SQL, MongoDB, PostgreSQL**) spanning all **3** production facilities, on a custom **Node.js/React/Redux/TypeScript** stack with **ETL/stream-processing** pipelines, automating entirely manual shop-floor workflows to lift operational efficiency **19%** and data reliability **14%**, and cutting **OEM/AFM** customer data-request turnaround by **90%**.
- Delivered end-to-end visibility via **Power BI** and custom dashboards showing real-time machine data, per-unit test results, and real-time **SPC**-based anomaly alerts — further strengthening the digital twin and powering the Digital Product Passport; accelerated reporting by **60%**, saving **\$156K** annually.

DVD VALVES — MANISA, TÜRKİYE

SOFTWARE DEVELOPMENT/AUTOMATION LEAD

Jan 2022 - Nov 2022

- Developed a **PLC-controlled** hydraulic butterfly valve control system with a **Modbus TCP-based SCADA** platform for international water projects (Italy; 8 × Ø700mm, Dubai; 2 × Ø1400/Ø1500mm, Colombia; 2 × Ø700mm); completed **FAT/SAT** testing. Brought development in-house instead of outsourcing, cutting costs by **30%** (**\$80k** saved).
- As project lead overseeing the **~\$3.5M TÜBİTAK TEYDEB 1501** budget, led a **3-person** core team (coordinating cross-departmental contributors across the project's **9+**-person team) building the embedded software and desktop/mobile applications for the pre-charge and ultrasonic metering modules of this smart irrigation hydrant system, cutting unit cost by **~68%** versus the outsourced solution and bringing after-sales technical support fully in-house.
- TÜBİTAK 2209-B** industrial consultant (smart agricultural valve project, No. 119B412103138): added **LoRa**-based remote meter reading.

SOFTWARE DEVELOPMENT/AUTOMATION ENGINEER

Jul 2018 - Dec 2021

- Designed two autonomous welding robots with **PLC/HMI** control to deliver a robust, high-quality weld in a sealing-critical zone, cutting welding cycle time by **60%** and reducing sealing-related quality defects by **23%**.
- Enabled real-time monitoring via the **Focas** library for **CNC** machines and **OPC UA** for **PLC**-based equipment; established **REST API**-based web-service data collection and historical data logging as part of this project.
- Modernized outdated, manually-operated control panels and revised **PLC** software for **10+** production-floor machines (blower test systems, sealing test units with an **MS SQL**-based desktop application, rubber press, sizing, polishing robot, saw, radial drill, and sandblasting), upgrading their automation level.
- Enabled real-time, cross-department visibility into production and maintenance operations through **MES** and **ERP** integration.

DİMES — İZMİR, TÜRKİYE

PART-TIME PROJECT ENGINEER

Jul 2017 - Jun 2018

- Transitioned a fully manual PET bottle production line to a fully autonomous conveyor system, designing the **PLC**-based control panel.
- Developed **PLC-SCADA** software for mixing tank control, enabling full traceability and digitizing reporting and quality processes, supporting faster detection and root-cause analysis of quality deviations.

Honors & Awards

Cevdet İnci Incentive Awards are held across İnci Holding's group companies (5 sister companies, 9+ facilities); İnci GS Yuasa Stars is company-specific.

2025	Finalist , Sustainability Business Awards (Technology & AI Category)	İstanbul, Türkiye
2025	1st Place , Cevdet İnci Incentive Awards	İzmir, Türkiye
2025	3rd Place , İnci GS Yuasa Stars	Manisa, Türkiye
2024	1st Place , Cevdet İnci Incentive Awards	İzmir, Türkiye
2024	Innovation Special Award , Cevdet İnci Incentive Awards	İzmir, Türkiye
2024	1st Place , İnci GS Yuasa Stars	Manisa, Türkiye
2023	1st Place , Cevdet İnci Incentive Awards	İzmir, Türkiye

Presentations & Publications

13th International European Conference on Interdisciplinary Scientific Research

[Tirana, Albania](#)

WRITER | RESEARCHER

May 2026

- Developed ML regression models (XGBoost, LightGBM, GBM) forecasting electricity and compressed air consumption on battery production lines, achieving **R² up to 0.944** for short-term forecasts.

R&D & Innovation 2024

[Manisa, Türkiye](#)

WRITER | RESEARCHER

Dec 2024

- Data Integration for Industry 4.0 and IIoT: Unified Namespace-Based Digital Transformation with **MQTT, OPC UA, and Node.js**.

Human-Computer Interaction Optimization and Robotic Applications

Türkiye

WRITER | RESEARCHER

Jun 2021

- Fault Detection and Diagnosis in Process Control Systems using **Machine Learning** (ensemble learning) Methods from Poincaré Plot Measurements, improving classification accuracy to **89.5%** on the Tennessee Eastman Process.

International Medical Device Conference

Antalya, Türkiye

WRITER | RESEARCHER

Sep 2020

- Long-Short Term Memory (LSTM)**-Based Heart Sounds Analysis and Classification: built an LSTM model classifying phonocardiogram recordings into Normal, Murmur, Extrasystole, and Artifact categories, achieving **79.0%** accuracy (Journal of Intelligent Systems and Applications, Vol. 3(1), 2020).

Certifications

41 NORTH BUSINESS SCHOOL

"GELECEĞİ SAHIPLENİYORUZ" BLACKBOX BUSINESS CHALLENGE™

Jun 2026

(Credential ID: INC_2026_0406_4118)

"GELECEĞİ SAHIPLENİYORUZ" CİDDİ OYUNLAR™ İLE GELECEĞİ SAHIPLENEN İNOVASYON PROJELERİ

Jul 2026

(Credential ID: INC_2026_2606_4106)

KALMER KALİTE YÖNETİM MERKEZİ

"ISO 9001:2015 & IATF 16949:2016" TRAINING

Feb 2023

(Document No: 39530)

N-SİSTEM

ISO 9001:2015, ISO 14001:2015, ISO 45001:2018

Nov 2021

(Credential ID: 2021/3994)

MANİSA YUNUSEMRE HALK EĞİTİM MERKEZİ

HYDRAULICS & PNEUMATICS DEVELOPMENT AND ADAPTATION TRAINING

Feb 2020

(64 hours | Document No: 75476720200065196147)

DATABASE MANAGEMENT – SQL DEVELOPMENT AND ADAPTATION TRAINING

Dec 2019

(56 hours | Document No: 75476720190063875731)

İZMİR HALK EĞİTİM MERKEZİ

COMPUTER-AIDED DESIGN (AUTOCAD)

Dec 2017

(160 hours, İzmir Büyükşehir Belediyesi | Document No: 16420920180053334601)

PROGRAMMABLE LOGIC CONTROL (PLC)

Dec 2017

(296 hours, İzmir Büyükşehir Belediyesi | Document No: 16420920170052946539)

Educations

İzmir Katip Çelebi University

İzmir, Türkiye

MSC. IN ELECTRICAL & ELECTRONICS ENGINEERING

Sep 2019 - Jun 2022

GPA: 3.64/4.00 (100% English-Medium Program).

Coursework: Statistical Process Control | Data Acquisition and Control | Applied Machine Learning | Artificial Neural Networks.

Thesis: Fault Detection and Diagnosis of the Tennessee Eastman Process Using Statistical Analysis and Machine Learning Methods from Poincaré Plot Measurements.

İzmir Katip Çelebi University

İzmir, Türkiye

BSC. IN ELECTRICAL & ELECTRONICS ENGINEERING

Sep 2014 - Jul 2019

GPA: 3.14/4.00 (100% English-Medium Program).

Coursework: Process Control and Instrumentation | Optimization in Engineering | Industrial Automation | Signals and Systems.

Thesis: Design of a Cylindrical Welding and Polishing Machine with PLC-Based HMI Control.